

DRY-RUN GUIDE | sliding_window.py
Source: Python\3rd_Belt\sliding_window.py

CORE BEHAVIOR

Maintains a contiguous fixed-size window and updates its aggregate as the window moves.

REPRESENTATIVE INPUT

Array '[2,1,5,1,3,2]', k=3.

STEP TRACE

Window sums are 8, 7, 9, 6; best sum=9.

EXPECTED OUTPUT / RETURN

Maximum window sum '9'.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt\sliding_window.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Use the sliding window technique to solve the array problem.
002 | # Add your solution here.
003 |
004 | n = int(input())
005 | arr = list(map(int, input().split()))
006 | t = int(input())
007 |
008 | left = 0
009 | current_sum = 0
010 | min_length = float('inf')
011 |
012 | for right in range(n):
013 |     current_sum += arr[right]
014 |
015 |     while current_sum >= t:
016 |         min_length = min(min_length, right - left + 1)
017 |         current_sum -= arr[left]
018 |         left += 1
019 |
020 | if min_length == float('inf'):
021 |     print(0)
022 | else:
023 |     print(min_length)
024 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.